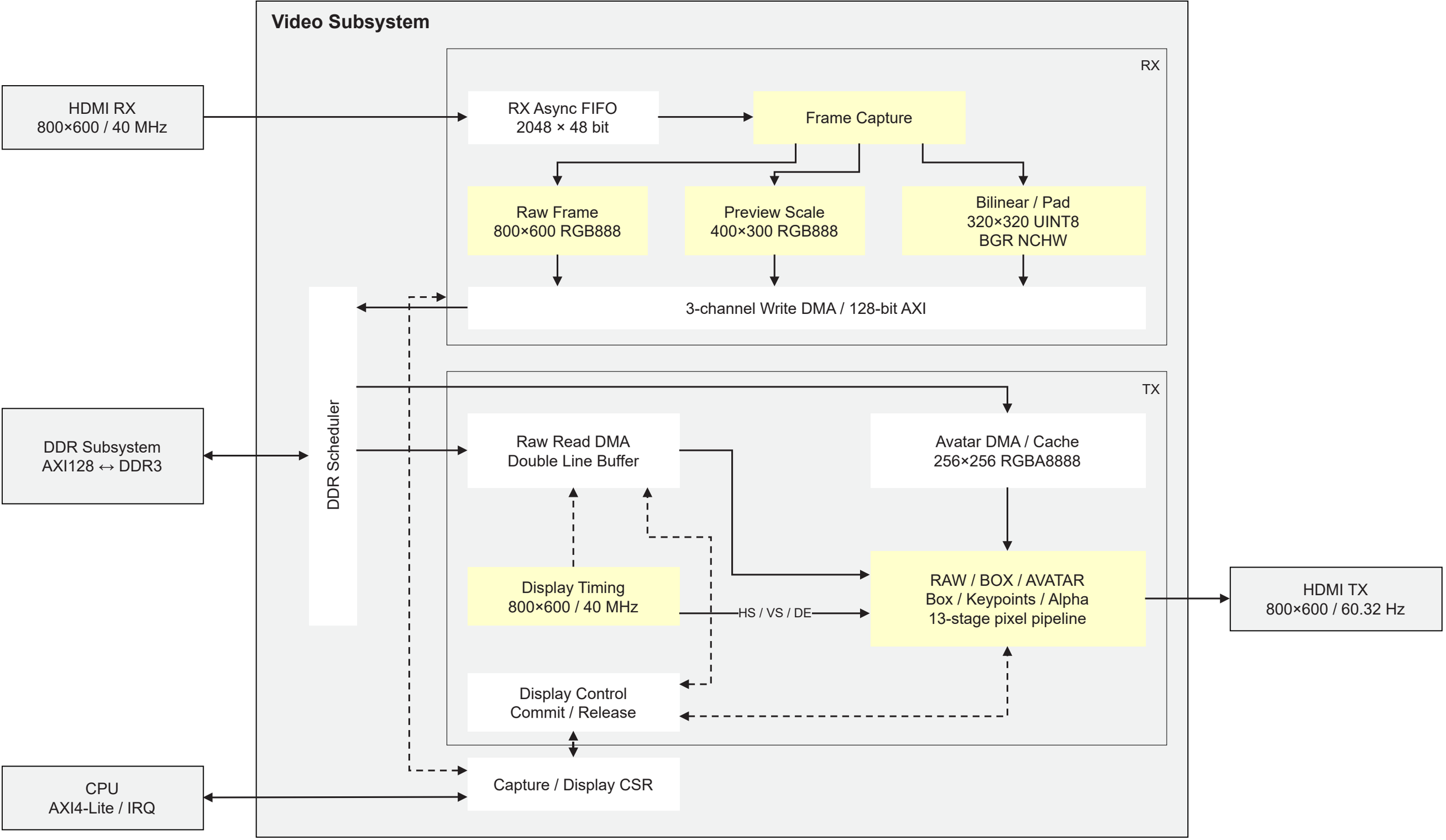
Video Subsystem
RX
HDMI RX
800×600 / 40 MHz
RX Async FIFO
2048 × 48 bit
Frame Capture
Raw Frame
800×600 RGB888
Preview Scale
400×300 RGB888
Bilinear / Pad
320×320 UINT8
BGR NCHW
3-channel Write DMA / 128-bit AXI
DDR Scheduler
TX
DDR Subsystem
AXI128 ↔ DDR3
Raw Read DMA
Double Line Buffer
Avatar DMA / Cache
256×256 RGBA8888
Display Timing
800×600 / 40 MHz
HS / VS / DE
RAW / BOX / AVATAR
Box / Keypoints / Alpha
13-stage pixel pipeline
HDMI TX
800×600 / 60.32 Hz
Display Control
Commit / Release
CPU
AXI4-Lite / IRQ
Capture / Display CSR